Homeostasis: Temperature Control

Exam code: 0610 & 0970

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Help me understand

The skin & homeostasis: extended

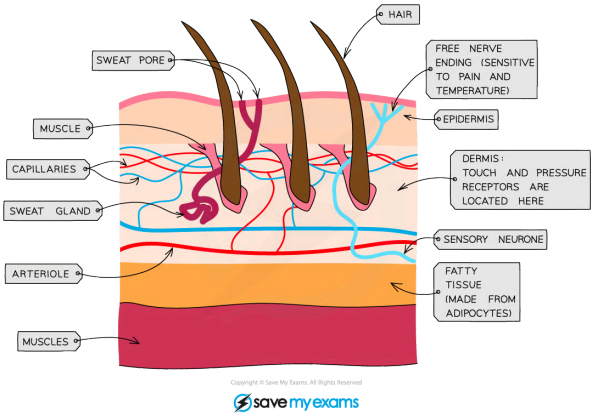
- Control of body temperature is a **homeostatic** mechanism
  - Homeostasis is **the maintenance of a constant internal environment**
  - This means that **internal conditions within the body** (such as temperature, blood pressure, water concentration and glucose concentration) need to be **kept within set limits** in order to ensure that reactions in body cells can function
- The human body maintains the temperature at which enzymes work best: around **37°C**
- If body temperature increases too far above this, **enzymes will denature** and become less effective at catalysing reactions

The skin and temperature regulation

- The **skin** plays an important role in the control of body temperature
- Structures in the skin that are involved with temperature regulation include:
  - sweat glands

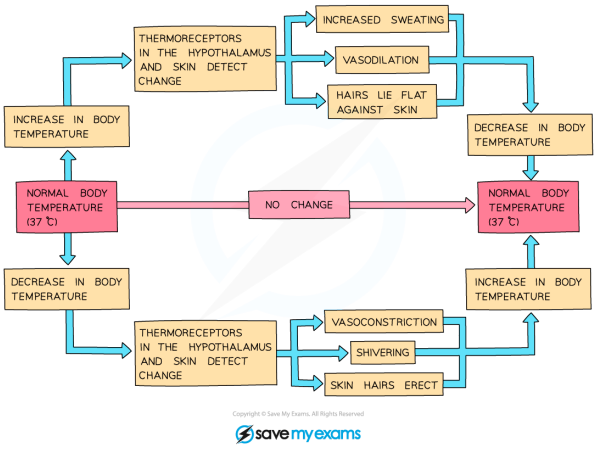
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- hair erector muscles
- insulating fatty tissue



The skin plays an important role in the control of body temperature

- When skin temperature changes, **temperature receptors** in the skin send nerve impulses to the brain via **sensory neurones**
- The brain responds to this information by sending nerve impulses to **effectors** in the skin that act to maintain normal body temperature, e.g.:
  - in response to an increase in body temperature:
    - **sweat glands** produce sweat which cools the skin by evaporation
    - blood vessels that supply the skin capillaries **dilate**
    - hairs on the skin surface **lie flat**
  - in response to a decrease in body temperature:
    - sweat gland stop producing sweat
    - blood vessels that supply the skin capillaries **constrict**
    - hairs on the skin stand up to **trap an insulating layer of air**
    - **shivering** generates heat in the muscles



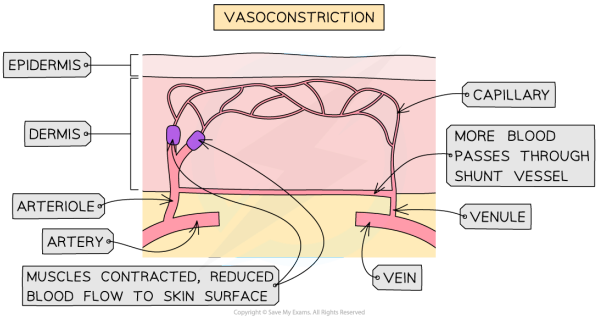
GLANDS THIS COOLS SKIN BY **EVAPORATION**. HEAT ENERGY FROM THE BODY IS LOST AS LIQUID WATER IN SWEAT BECOMES WATER VAPOUR (A STATE CHANGE).	AND WE **SHIVER**. THESE INVOLUTARY MUSCLE CONTRACTIONS NEED **ENERGY FROM RESPIRATION** AND SOME OF THIS IS **RELEASED AS HEAT**.
HAIRS **LIE FLAT** AGAINST THE SKIN, ALLOWING AIR TO FREELY CIRCULATE. THIS INCREASES HEAT TRANSFER TO ENVIRONMENT BY **RADIATION**.	ERECT HAIRS **TRAP A LAYER OF AIR AROUND THE SKIN** WHICH ACTS AS AN **INSULATOR**, PREVENTING HEAT LOSS BY RADIATION.

Vasoconstriction & vasodilation: extended

- The blood vessels in and below the skin aid temperature regulation by vasoconstriction and vasodilation

Vasoconstriction

- When we are **cold** the blood vessels that supply capillaries in the skin constrict, or get narrower; this is known as **vasoconstriction**
- Vasoconstriction reduces blood flow to the skin, and so **reduces the heat lost from blood by radiation** at the skin surface



Vasoconstriction reduces blood flow to the skin, and so reduces the heat lost from blood by radiation

Vasodilation

- When we are **hot** the blood vessels that supply capillaries in the skin dilate, or get wider; this is known as **vasodilation**
- Vasodilation increases blood flow to the skin surface, and so **increases heat lost at the skin by radiation**

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